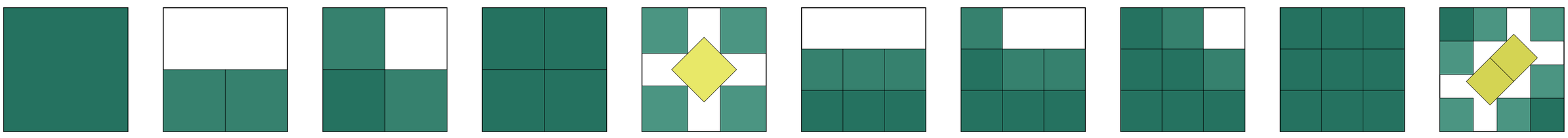
100 BEST KNOWN SQUARE PACKINGS

The Squares Project · github.com/jlevy/squares



1 O = R
s(1) = 1

2 O =
s(2) = 2

3 O =
s(3) = 2

4 O = R
s(4) = 2

5 O = R
s(5) = 2.707107 deg 2

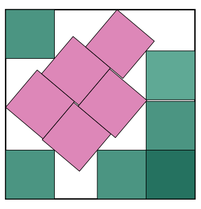
6 O =
s(6) = 3

7 O =
s(7) = 3

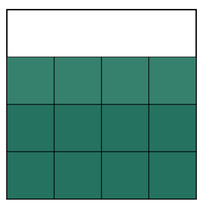
8 O =
s(8) = 3

9 O = R
s(9) = 3

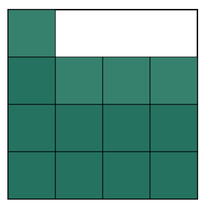
10 O =
s(10) = 3.707107 deg 2



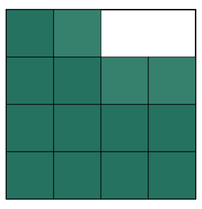
11 O = R
s(11) = 3.877084 deg 8



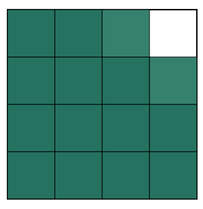
12 =
s(12) ≤ 4
s(12) ≥ 3.971500



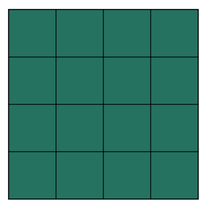
13 O =
s(13) = 4



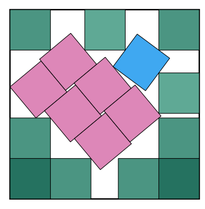
14 O =
s(14) = 4



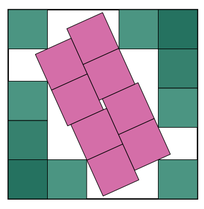
15 O =
s(15) = 4



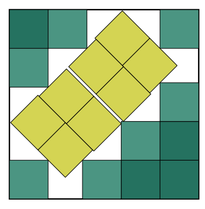
16 O = R
s(16) = 4



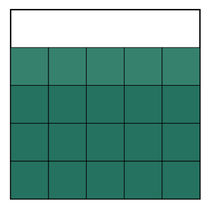
17 =
s(17) ≤ 4.675531 deg 18
s(17) ≥ 4.660442



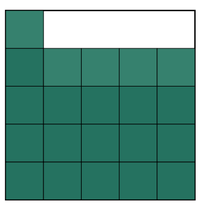
18 =
s(18) ≤ 4.822876 deg 2
s(18) ≥ 4.705000



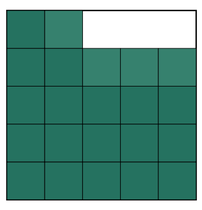
19 =
s(19) ≤ 4.885619 deg 2
s(19) ≥ 4.825000



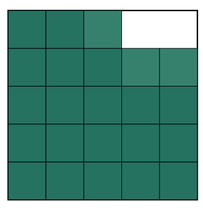
20 =
s(20) ≤ 5
s(20) ≥ 4.905000



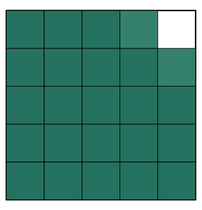
21 O =
s(21) = 5



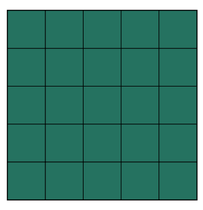
22 O =
s(22) = 5



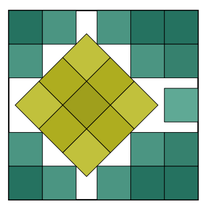
23 O =
s(23) = 5



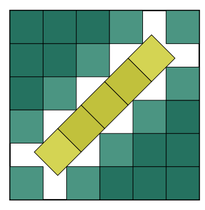
24 O =
s(24) = 5



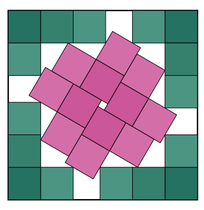
25 O = R
s(25) = 5



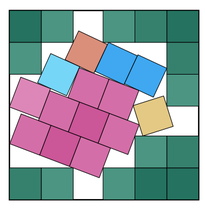
26 =
s(26) ≤ 5.621321 deg 2
s(26) ≥ 5.545000



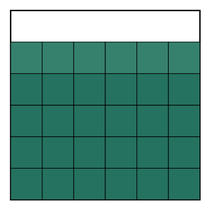
27 =
s(27) ≤ 5.707107 deg 2
s(27) ≥ 5.643500



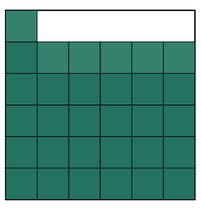
28 = R
s(28) ≤ 5.824445 deg 6
s(28) ≥ 5.735000



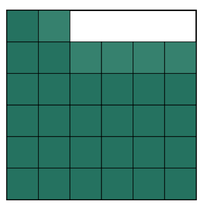
29 ≈
s(29) ≤ 5.933834
s(29) ≥ 5.810000



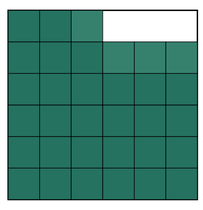
30 =
s(30) ≤ 6
s(30) ≥ 5.883500



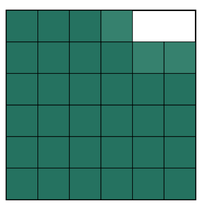
31 =
s(31) ≤ 6
s(31) ≥ 5.952500



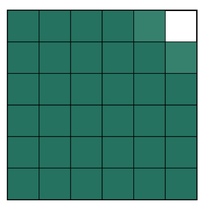
32 O =
s(32) = 6



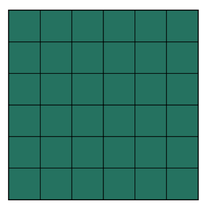
33 O =
s(33) = 6



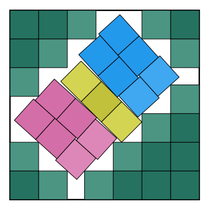
34 O =
s(34) = 6



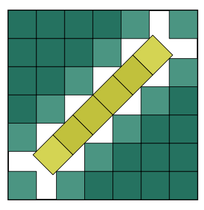
35 O =
s(35) = 6



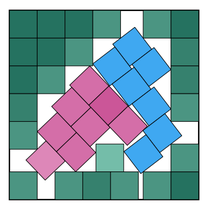
36 O = R
s(36) = 6



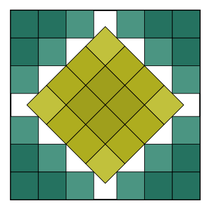
37 =
s(37) ≤ 6.598620 deg 8
s(37) ≥ 6.440000



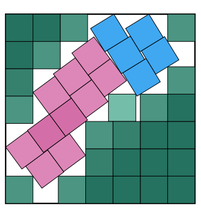
38 =
s(38) ≤ 6.707107 deg 2
s(38) ≥ 6.545000



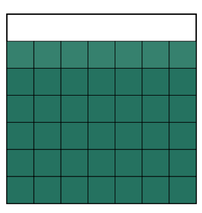
39 =
s(39) ≤ 6.810723 deg 5
s(39) ≥ 6.650000



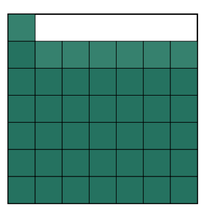
40 = R
s(40) ≤ 6.828428 deg 2
s(40) ≥ 6.700000



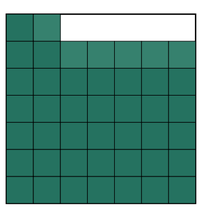
41 =
s(41) ≤ 6.926694 deg 42
s(41) ≥ 6.775000



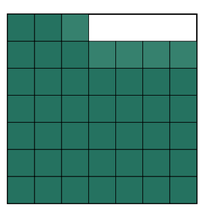
42 =
s(42) ≤ 7
s(42) ≥ 6.847500



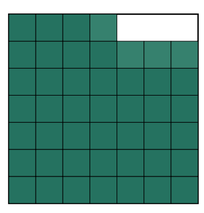
43 =
s(43) ≤ 7
s(43) ≥ 6.907500



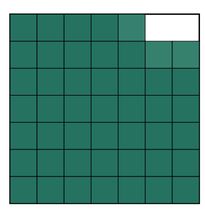
44 =
s(44) ≤ 7
s(44) ≥ 6.972500



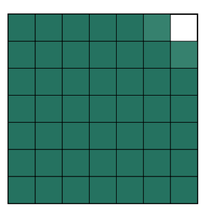
45 O =
s(45) = 7



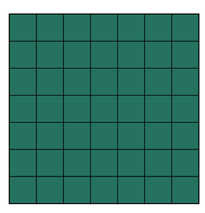
46 O =
s(46) = 7



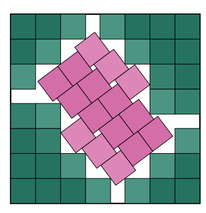
47 O =
s(47) = 7



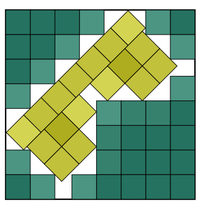
48 O =
s(48) = 7



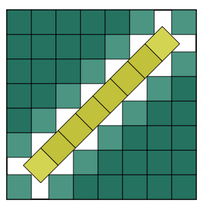
49 O = R
s(49) = 7



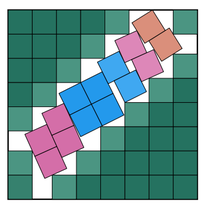
50 =
s(50) ≤ 7.571429
s(50) ≥ 7.400000



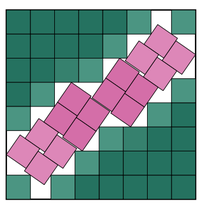
51 =
s(51) ≤ 7.690356 deg 2
s(51) ≥ 7.470000



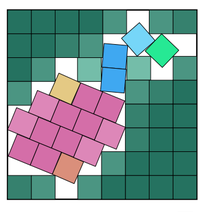
52 =
s(52) ≤ 7.707107 deg 2
s(52) ≥ 7.550000



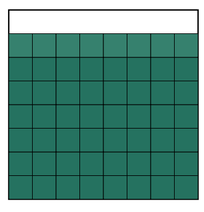
53 =
s(53) ≤ 7.822876 deg 2
s(53) ≥ 7.627500



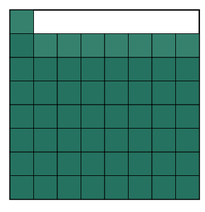
54 =
s(54) ≤ 7.846668 deg 4
s(54) ≥ 7.685000



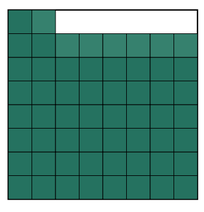
55 ≈
s(55) ≤ 7.945772
s(55) ≥ 7.728000



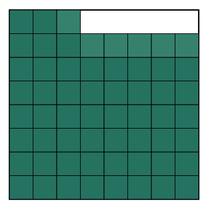
56 =
s(56) ≤ 8
s(56) ≥ 7.802500



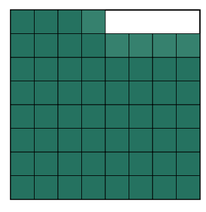
57 =
s(57) ≤ 8
s(57) ≥ 7.872500



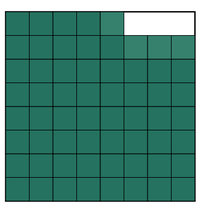
58 =
s(58) ≤ 8
s(58) ≥ 7.935000



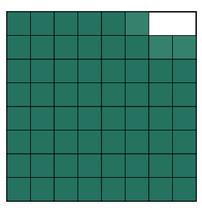
59 O =
s(59) = 8



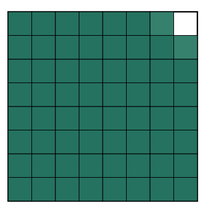
60 O =
s(60) = 8



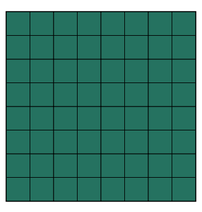
61 O =
s(61) = 8



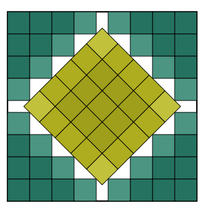
62 O =
s(62) = 8



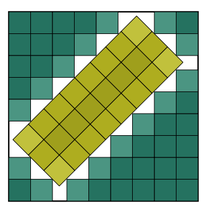
63 O =
s(63) = 8



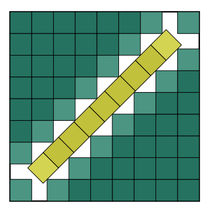
64 O = R
s(64) = 8



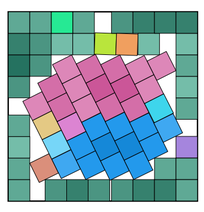
65 =
s(65) ≤ 8.535534 deg 2
s(65) ≥ 8.350000



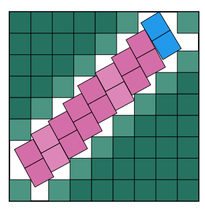
66 =
s(66) ≤ 8.656855 deg 2
s(66) ≥ 8.430000



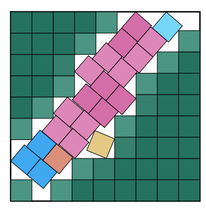
67 =
s(67) ≤ 8.707107 deg 2
s(67) ≥ 8.480000



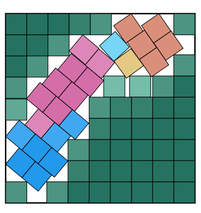
68 =
s(68) ≤ 8.798796
s(68) ≥ 8.510000



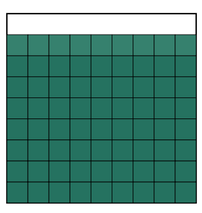
69 =
s(69) ≤ 8.827195 deg 38
s(69) ≥ 8.620000



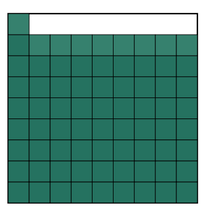
70 =
s(70) ≤ 8.880961
s(70) ≥ 8.657500



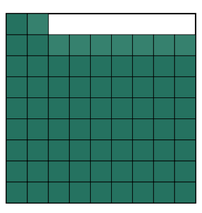
71 ≈
s(71) ≤ 8.944072
s(71) ≥ 8.721000



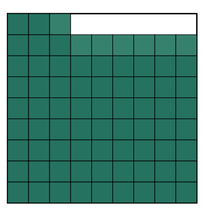
72 =
s(72) ≤ 9
s(72) ≥ 8.760000



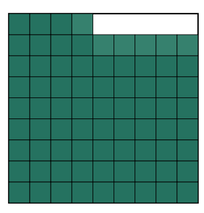
73 =
s(73) ≤ 9
s(73) ≥ 8.813000



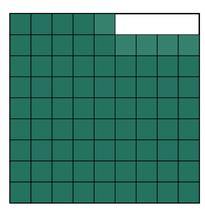
74 =
s(74) ≤ 9
s(74) ≥ 8.867500



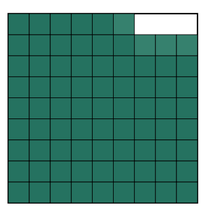
75 =
s(75) ≤ 9
s(75) ≥ 8.940000



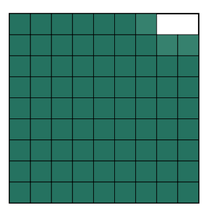
76 =
s(76) ≤ 9
s(76) ≥ 8.965000



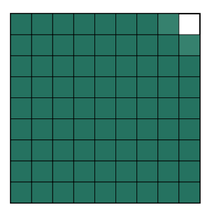
77 O =
s(77) = 9



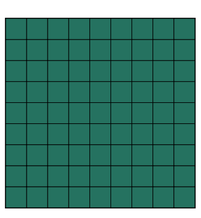
78 O =
s(78) = 9



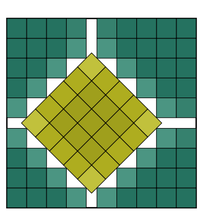
79 O =
s(79) = 9



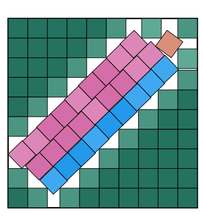
80 O =
s(80) = 9



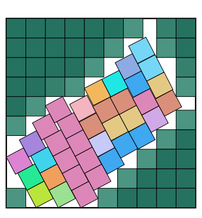
81 O = R
s(81) = 9



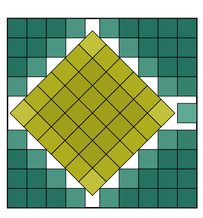
82 =
s(82) ≤ 9.535534 deg 2
s(82) ≥ 9.320000



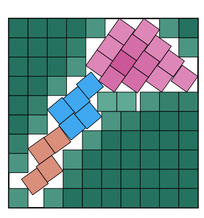
83 =
s(83) ≤ 9.634758deg 672
s(83) ≥ 9.370000



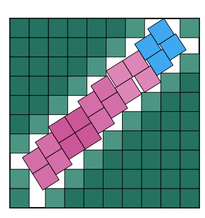
84 =
s(84) ≤ 9.698053
s(84) ≥ 9.411000



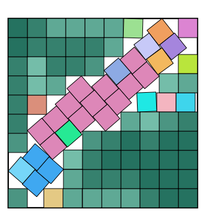
85 =
s(85) ≤ 9.742641 deg 2
s(85) ≥ 9.460000



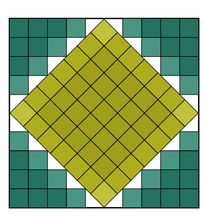
86 =
s(86) ≤ 9.820566
s(86) ≥ 9.503000



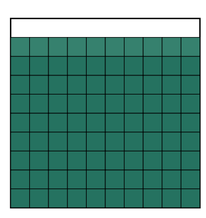
87 =
s(87) ≤ 9.838816 deg 41
s(87) ≥ 9.580000



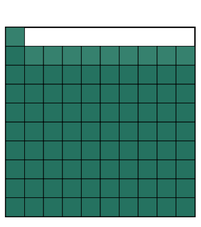
88 =
s(88) ≤ 9.882452
s(88) ≥ 9.620000



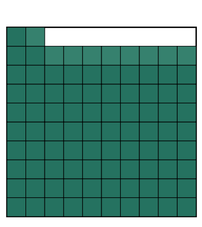
89 =
s(89) ≤ 9.949748 deg 2
s(89) ≥ 9.650000



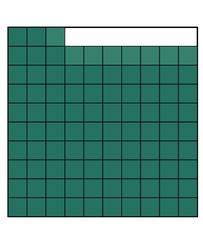
90 =
s(90) ≤ 10
s(90) ≥ 9.730000



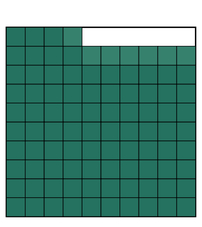
91 =
s(91) ≤ 10
s(91) ≥ 9.762500



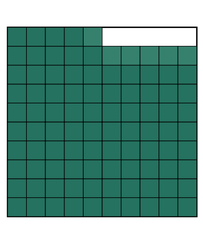
92 =
s(92) ≤ 10
s(92) ≥ 9.770000



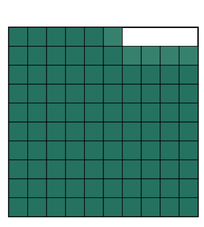
93 =
s(93) ≤ 10
s(93) ≥ 9.880000



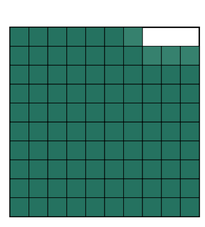
94 =
s(94) ≤ 10
s(94) ≥ 9.950000



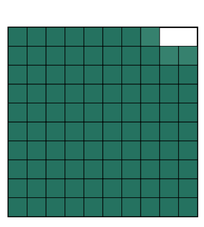
95 =
s(95) ≤ 10
s(95) ≥ 9.965000



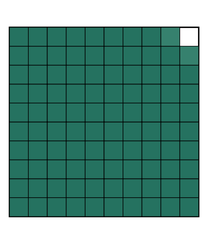
96 =
s(96) ≤ 10
s(96) ≥ 9.970000



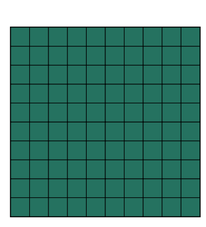
97 O =
s(97) = 10



98 O =
s(98) = 10



99 O =
s(99) = 10



100 O = R
s(100) = 10

The square packing problem asks for the side *s*(*n*) of the smallest square that can hold *n* unit squares, where the squares are free to rotate but cannot overlap

O proved optimal (45 of 100)
= exact value known (97 of 100)
≈ only known numerically (3 of 100)
R rigid (14 of 100)

90° 45° colors indicate distinct tilt angles
4 3 2 1 0 shade indicates number of full-side contacts
deg is the algebraic degree of that side length
recent result (since August, 2026)

Best packings due to Göbel, Trump, Wainwright, Hämäläinen, Stenlund, Friedman, Bidwell, Cantrell, Morandi, Ellsworth, Hajba, Schadt, Chaoweeraprasit, Couzo, Daniel, de Winter, Rehwaldt, Gupta, ry-xu, Chang, DeVincentis, hmbelvedere

Lower bounds due to Göbel, Stromquist, Kearney, Shiu, Bentz, Daniel, Vlasenko, Guzhou0806, Hosono, Ahmed, Karakuş, Ryu

Optimality proofs due to Göbel, Stromquist, Friedman, El Moumni, Kearney, Shiu, Bentz, Vlasenko, Levy, Daniel, Hosono, Ahmed, Karakuş; formalization contributions by Gurevitch, Guzhou0806; verification infrastructure and execution by EvolvingPrograms, Julian-JJ, Lewis

Diagram by Joshua Levy
October 10, 2026 · v0.5.0-c8abd5